T'Shara Haynes

Software Engineer | Research | Development

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PROFESSIONAL SUMMARY

I'm a software engineer who enjoys building thoughtful digital experiences backed by strong technical systems. I'm particul arly interested in machine learning research, game development, social media marketing, and designing tools to further th e Caribbean's technological and sustainable development.

WORK EXPERIENCE

Tessera Studios – Remote, Barbados
Contract Project Assistant **February 2026 – Present**

- Managed and scheduled social media content across multiple platforms to ensure consistent and timely publicatio n.
- Wrote engaging copy for social media captions and blog content while maintaining brand voice and clarity.
- Designed visual assets including social media posts, carousels and stories.
- Edited video content for social media platforms, incorporating captions, music, and transitions to enhance audienc e engagement.

Central Bank of Barbados - Bridgetown, Barbados
Governor's Office Intern **June 2025- August 2025**

- Mapped end-to-end payment system workflows using system documentation to support integration for the BiMPay rollout.
- Authored a research paper on fraud detection in instant payment systems, presented at the Annual Review Seminar.
- Produced clear process documentation to enable cross-team implementation.
- Reviewed regional data protection standards to ensure applicability of compliant use of AI and customer data.
- Developed fraud detection models to analyse transaction data and strengthen risk monitoring.

Pellucid IT Inc - Remote, Barbados
Project Assistant **October 2024- May 2025**

- Documented team workflows and improved task tracking to keep projects organized and on schedule.
- Implemented Scrum Methodology to enhance project agility and streamline workflows.
- Tracked timelines, identified blockers, and kept stakeholders informed to ensure smooth project delivery.

Headstarter AI - Remote, Trinidad and Tobago
Software Engineering Fellow **July 2024- September 2024**

- Created an inventory tracking application with React and Next.js to manage household supplies
- Developed a responsive portfolio website using HTML, CSS, and JavaScript to showcase technical projects
- Integrated OpenAI API with Next.JS to generate AI-powered study flashcards.

EDUCATION

University of the West Indies, Cave Hill Campus
Bachelor of Science in Computer Science w/ Environmental Science **Expected Graduation: May 2026**

SKILLS & INTERESTS

Programming Languages - Python, SQL, Java, PHP, JavaScript, HTML/CSS
Databases: MySQL, MongoDB
Frameworks/ Tools – NodeJS, Git, React, Streamlit, Bash, Pandas
Design Tools – Canva, Figma, Lightroom, Gimp, Aseprite
Methods– Technical Documentation, Agile/Scrum

PROJECTS

Fraud Detection Simulator **Research Project**
Python | Streamlit | SciKit-Learn

- Using Python and Streamlit, built a real-time transaction simulation app to model IPS payment flows.
- Applied Scikit-learn feature engineering and model evaluation techniques to detect anomalies in transaction data (~92% accuracy)
- Built Python-based simulation pipeline to ingest and analyse streaming transaction data in real-time
- Measured classification accuracy, precision-recall, ROC curves, and average detection time for each approach.
- Focused on challenges in low-label Caribbean IPS environments.

Energy Dashboard **Personal Project**
Python| Streamlit

- Developed a multi-building energy consumption dashboard with interactive visualisations, KPI metrics and scenari o simulation for usage reduction modelling.
- Implemented time-series forecasting using Facebook Prophet with custom date filtering and CSV functionality.

Sign Language Text-to-Speech Detector **Research Project**
Python| Tkinter | OpenCV

- Trained a machine learning model using Google's Teachable machine to classify sign language gestures.
- Pre-processed and labelled dataset; performed feature extraction to improve accuracy.
- Implemented real-time interference pipeline using OpenCV and Python.
- Designed and built an interactive Tkinter GUI that displayed signs as text and speech output.

RESEARCH & PUBLICATIONS

- Fraud Detection in Instant Payment Systems: A Caribbean Case Study of Machine Learning Constraints and Opportunities (2025).
- Augmentation of Machine Learning Algorithms for Air Quality Predictions (2026)
- Artificial Intelligence in Sign Language Text-to-Speech (2025)

LEADERSHIP

UWI Debate Society | University of the West Indies Cave Hill, Bridgetown, Barbados
Secretary **June 2024 - October 2025**

- Organised meetings facilitated communication between members and assisted in event planning and execution.
- Created an "Intro to Debate" booklet to engage with members and provide debating information to upcoming members and executive

Faculty of Science and Technology Representative | University of the West Indies Cave Hill, Bridgetown, Barbados *Computer Science Representative* **June 2024 – May 2025**

- Acted as a liaison between students and faculty, addressed concerns, and coordinated academic and extracurricular initiatives for Computer Science students

Trinidad and Tobago Youth Advocacy Network| ***Trinidad and Tobago***
Environmental Department Member **October 2024 – Present**

- Engaged in advocacy initiatives focused on youth empowerment, policy change, and social impact.
- Collaborated with like-minded individuals and organizations to promote sustainable development goals (SDGs).

ADDITIONAL CERTIFICATES

AWS:

- Getting Started with Security
- Introduction to Cloud 101

Coursera:

- DevOps for Network Automation (NetDevOps)
- Google Project Management

REFERENCES

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